

Three Ways to Assign Samples to Batches

10 cases (C) + 10 controls (K) across 2 batches

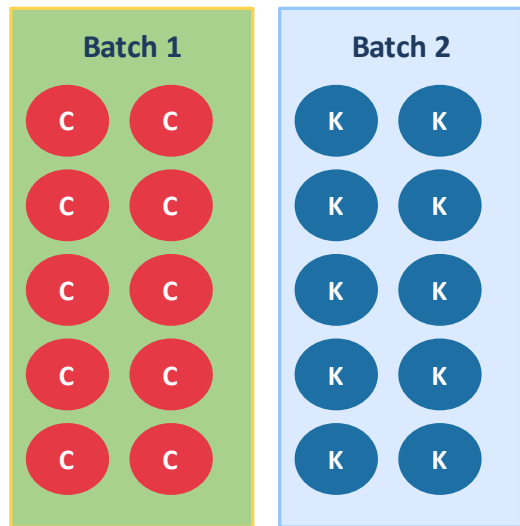


Case



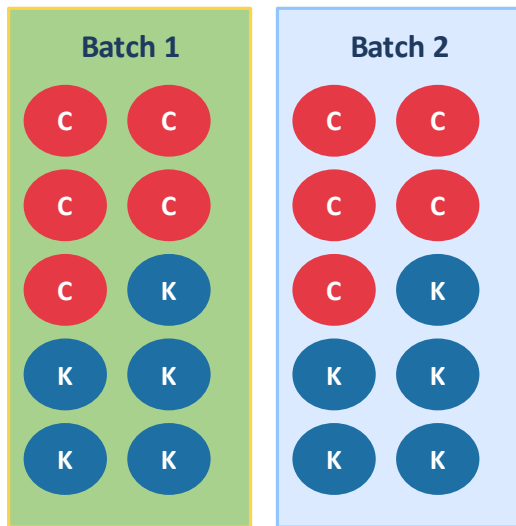
Control

✗ Confounded



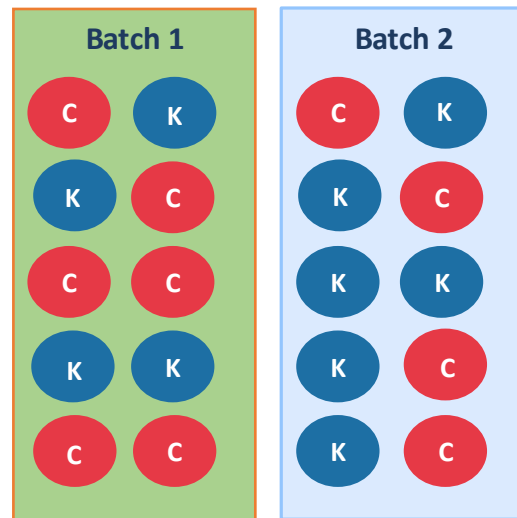
*Batch $\equiv$ Biology
Cannot be separated*

Blocked



*Batch estimable & removable
from within-batch contrasts*

Randomised



*Chance balance
Reduces systematic bias*

Comparison: All Three Strategies

Property	✗ Confounded	Blocked	Randomised
Batch $\leftrightarrow$ Biology	Inseparable	Separated by design	Partially separated
Batch estimable?	No	Yes	Sometimes
Statistical model	Cannot adjust	$\sim$ condition + batch	May adjust post-hoc
False positives	High risk	Controlled	Reduced
Power	Reduced (confound)	Maximised	Good
When to use	Never (avoid)	Preferred strategy	When blocking impractical